Vital Signs: Hospital Finance

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GitHub link

Introduction

Hospital financial instability has become an increasingly important policy issue in the United States, particularly in the years surrounding the COVID-19 pandemic. Federal relief funding temporarily strengthened reported hospital finances, but the expiration of those programs revealed renewed financial strain for many institutions. This raises a critical question: can hospitals at risk of persistent financial distress be identified using publicly available administrative data before closure or severe service reductions occur?

Our team wanted to tackle this problem because hospital closures severely reduce access to care. Hearing about these closures made us want to see if the data could have warned us. If we can accurately predict which hospitals are failing early on, policymakers and health administrators could step in with targeted funding or operational help before a community completely loses its hospital.

To build our models, we used publicly available CMS Hospital Cost Reports and Provider of Services data spanning from 2015 to 2023. We deliberately split our timeline to test how well the models generalize across massive economic disruptions. We trained our supervised models on the pre pandemic years from 2017 to 2019 and used 2022 to 2023 as our out of sample test period. For supervised learning, we defined distress strictly as a hospital experiencing two consecutive years of negative operating margins while also falling into the bottom ten percent for liquidity. In our dataset, we labeled this distress B because it signals a persistent structural problem rather than a single bad year. We then trained Logistic Regression, Support Vector Classifiers, and Gradient Boosting models to predict this outcome.

For unsupervised learning, we wanted to see if hospitals naturally grouped into distinct risk profiles. We aggregated the features to create hospital level averages and applied Principal Component Analysis alongside KMeans and Gaussian Mixture Models. What makes our approach novel is that we actively prevented data leakage by rigorously scrubbing variables like Net Income and Total Current Assets that were directly used to calculate our distress label. This forced the model to learn true predictive operational patterns instead of just reading the answers. Furthermore, we introduced a novel refinement in Version B of our unsupervised clustering phase. Because our initial PCA showed clustering was heavily skewed by extreme financial outliers, we clipped all numeric features at the 1st and 99th percentiles and applied a RobustScaler. This ensured our final segments reflected actual hospital characteristics rather than extreme anomalies.

We extracted massive insights from both halves of our project. For supervised learning, Gradient Boosting completely outperformed the linear models, proving that hospital distress patterns are highly nonlinear. The tuned Gradient Boosting model generalized beautifully on our test set, achieving an ROC AUC of 0.886 and a PR AUC of 0.395. Our ablation analysis also proved that financial ratios do the heavy lifting for predictions. When we removed financial features, the cross validation PR AUC mean plummeted from 0.508 down to 0.195. For unsupervised learning, our clustering algorithms found underlying structure, with Silhouette scores suggesting KMeans worked best with 2 clusters, while Bayesian Information Criterion scores optimized our Gaussian Mixture Model at 7 components. However, clustering hospital data is incredibly sensitive to feature scaling. When we compared our RobustScaler assignments against a StandardScaler approach, the Adjusted Rand Index was negative 0.024, showing total disagreement between the methods. Additionally, our failure analysis showed that while the vast majority of hospitals form stable clusters, 53 facilities consistently presented outlier-like profiles and 53 exhibited low confidence method uncertainty. Our results prove that persistent financial vulnerability can be detected early using public data, and combining predictive modeling with structural clustering offers a highly effective early warning system.

Related Work

Three examples of existing studies that we found that were similar to our project were relevant, but we actively worked to improve upon their methods in our own analysis.

First, we reviewed the study titled Rural Hospital Financial Risk (available at https://pubmed.ncbi.nlm.nih.gov/26171848/), which examined the predictors of rural hospital closures using financial indicators from medicare cost reports and structural hospital characteristics. The authors used logistic regression models to identify financial and market factors associated with elevated closure risk. While this study mainly used logistic regression and looked at a single snapshot of the past, we wanted to see what happens when we use more advanced models like gradient boosting. We also wanted to test if these models could actually predict distress across a major economic shift instead of just memorizing data from one normal time period.

Second, to broaden our scope beyond rural facilities, we found a study applying advanced algorithms to national healthcare datasets titled Financial Distress Prediction for US Hospitals with Machine Learning (available at https://norma.ncirl.ie/7307/1/manelhmida.pdf). This study shows how classification models can identify vulnerable institutions using operational ratios. While their machine learning application focuses solely on binary prediction, we improve upon their foundation by integrating unsupervised clustering to segment structural financial profiles before applying our predictive models. This ensures our framework not only predicts distress but also identifies hidden heterogeneity in hospital risk that a single static risk threshold might miss.

Third, we analyzed the study COVID-19 and the Financial Health of US Hospitals (available at https://jamanetwork.com/journals/jama/fullarticle/2765698), which examined the impact of pandemic relief funding on hospital operating margins. The research showed that federal relief funds temporarily offset large revenue declines in 2020, and the analysis highlighted how pandemic era support masked underlying financial stress for some hospitals. While their study focused on the immediate financial impact and the masking effect of pandemic relief, our project specifically evaluates out of sample prediction using a post pandemic test period. This allows us to assess whether pre-pandemic structural data can reliably identify renewed vulnerability after those relief funds expire.

Data Sources

In this project, we used two publicly available datasets from the Centers for Medicare and Medicaid Services (CMS): Hospital Cost Reports (HCRIS) and Provider of Services (POS) Files. Together, these datasets were merged to create a multi-year hospital-level dataset containing financial, operational, and structural characteristics. The combined dataset was used for both supervised prediction financial distress and unsupervised clustering analysis.

CMS Hospital Cost Reports (HCRIS)

- Location: The Hospital Cost Reports were downloaded from the CMS public data portal: https://data.cms.gov/provider-compliance/cost-reports/hospital-provider-cost-report/data
- Format: The data are provided as annual CSV datasets. We used data from 2015-2023, although the modeling period focuses on 2017-2023.
- Important Variables:
    - Key financial variables: net income, net patient revenue, total current assets, total current liabilities, total assets, total liabilities, total costs, costs of uncompensated care.
    - Operational variables: total discharges, total days
- Records used or retrieved: After cleaning and merging duplicate filings, the final supervised modeling dataset included hospital-year observations from 2017–2023. The held-out test set (2022–2023) contains 9,643 hospital-year observations.
- Time Period Covered: We used data from 2015-2023, although the modeling period focuses on 2017-2023.

With these fields, financial and utilization ratios were calculated, including operating margin, liquidity, leverage, occupancy rate, and uncompensated care ratio. These variables were used to construct the binary outcome variable for financial distress.

CMS Provider of Services (POS) Files

- Location: The Provider of Services files were downloaded from: https://data.cms.gov/provider-characteristics/hospitals-and-other-facilities/provider-of-services-file-quality-improvement-and-evaluation-system/data
- Format: Annual CSV files (Q4 releases). Because the files are large, they were read in chunks during preprocessing.
- Important Variables:
    - Structural and geographic variables: hospital subtype and provider category, FIPS state code, urban vs. rural indicator, CBSA (metropolitan area) code, certified bed count, medicare/medicaid participation status.
- Records used or retrieved: the POS files provide structural and geographic characteristics. Because of file size, data were loaded in chunks and filtered to include only hospitals appearing in the Cost Report dataset.
- Time Period Covered: 2015-2023

The variables selected were used as predictors in the supervised models and as inputs for clustering in the unsupervised analysis.

After merging the two datasets on Provider CCN and year, the final supervised dataset consists of hospital-year observations between 2017-2023. For the unsupervised analysis, we limited the data to the pre-pandemic period (2017–2019) and aggregated the data to one row per hospital by averaging key financial and operational features. This produced 5,028 unique hospitals initially, with approximately 1,050 hospitals remaining after filtering out those without complete clustering features. The resulting dataset integrates financial performance, utilization metrics, and structural characteristics, providing the foundation for analyzing and predicting hospital financial distress.

Feature Engineering

We started by merging two public CMS datasets from 2015 to 2023: the Hospital Cost Reports and the Provider of Services files. A major issue we had to clean up first was that some hospitals submitted multiple cost reports in a single fiscal year. To fix this, we sorted the filings by the fiscal year end date to keep the most recent record, but we summed important volume and financial metrics like total discharges, inpatient revenue, and total costs across the duplicate filings before merging.

Next, we used the raw data to calculate the financial and operational ratios needed for our models. We used the following formulas:

- Operating Margin: net income from service to patients divided by net patient revenue.
- Net Margin: net income divided by net patient revenue.
- Liquidity: total current assets divided by total current liabilities.
- Leverage: total liabilities divided by total assets.
- Average Length of Stay: total days divided by total discharges.
- Occupancy Rate: total days divided by the product of certified bed count and 365.
- Payor Mix: net revenue from Medicaid divided by net patient revenue.
- Uncompensated Care Ratio: cost of uncompensated care divided by total costs.
- Resident Bed Ratio: number of interns and residents divided by certified bed count.

We learned that dividing by zero sometimes created infinite values, so we replaced all infinite values with null values. Since operating margin and liquidity were strictly required to calculate our financial distress target variable, we dropped any hospital rows that were missing those two metrics.

When building our supervised machine learning features, we applied a major concept from our course material regarding data leakage. We knew the model could not have access to the information it was trying to predict, so we removed the raw variables used to calculate the distress label from our predictor set. This meant dropping net income, total current assets, total current liabilities, operating margin, and liquidity. We also applied course concepts for handling heterogeneous data by building a scikit learn ColumnTransformer. We passed numeric features through a SimpleImputer using the median strategy and a StandardScaler, while categorical features went through a SimpleImputer and a One Hot Encoder.

Our final supervised features were grouped into three categories to represent different dimensions of hospital performance:

- Numeric Financial: net margin, leverage, payor mix, uncompensated care ratio, and resident bed ratio. These capture profitability and capital structure risk.
- Numeric Utilization: average length of stay, occupancy rate, total discharges, total days, and certified bed count. These measure capacity intensity and operational volume.
- Categorical Structural: provider category subtype code, provider category code, state FIPS code, CBSA urban rural indicator, and program participation code. These show geographic variations and regulatory status.

For our unsupervised clustering models, we needed to evaluate structural profiles instead of annual performance. We filtered the data to the pre pandemic years of 2017 to 2019 and combined it into a single row per hospital. We did this by taking the mean of the numeric features and the mode of the categorical features. During our first clustering attempt, we realized that extreme financial outliers were heavily skewing the principal component analysis. To fix this in Version B of our clustering model, we clipped all numeric features at the 1st and 99th percentiles. We also swapped the standard scaler for a RobustScaler to make sure the clusters grouped actual hospital characteristics rather than extreme anomalies.

Our final unsupervised features for the refined model included:

- Numeric: operating margin, liquidity, leverage, occupancy rate, certified bed count, payor mix, uncompensated care ratio, and resident bed ratio.
- Categorical: CBSA urban rural indicator and provider category code.

Part A: Supervised Learning

Supervised Methods Description

Our supervised learning objective was to build a binary clarifier that predicts hospital financial distress using publicly available CMS Cost Report and Provider of Services data. We defined our target label as persistent financial distress (distress_B) defined by hospitals experiencing two consecutive years of negative operating margin and falling in the bottom 10% of liquidity based on pre-pandemic years. We intentionally chose this definition because it captures structural financial instability rather than a single year fluctuation (distress_A). This aligns with our larger goal of identifying hospitals at sustained risk rather than reacting to temporary shifts.

In order to avoid data leakage, we removed any variables directly used in constructing the distress label (operating margin, liquidity, net income, total current assets/liabilities, etc.). This forces the model to learn predictive patterns from independent financial and structural variables rather than reconstructing the target variable.

We used a temporal split to imitate real-world deployment, training was conducted on years 2017-2019 and testing on more recent years 2022-2023. We intentionally excluded the COVID-19 period from model training due to government relief funding that impacted hospital financial performance, sudden shifts in elective procedures, and

abnormal utilization patterns. Including COVID-19 years in the training model would risk teaching the model patterns that were driven by temporary policy interventions rather than underlying financial vulnerability. We were able to better assess whether the model captures structural distress signals that generalize beyond extraordinary shocks by training on stable years and evaluating on more recent data.

Feature Representation

We created financial, operational, and structural features from CMS Cost Report and Provider of Services data to capture different dimensions of hospital performance. Our goal was to represent both financial and operational strain.

Financial Structure and Stability

The following financial ratios were created from the CMS Cost Report:

- Operating margin = Net Income from Service of Patients / Net Patient Revenue
- Liquidity = Total Current Assets / Total Current Liabilities
- Net margin = Net Income / Net Patient Revenue
- Leverage = Total Liabilities / Total Assets

Net margin, as defined in our implementation, reflects profitability relative to patient generated revenue rather than total revenue. This focuses the signal on core clinical operations rather than auxiliary or non-operating income streams. Leverage captures capital structure risk and long-term financial obligations compared to asset base which is used as a financial fragility proxy. Operating margin reflects performance from operating activities and liquidity measures short-term solvency.

Operational Metrics

The following utilization were both created or pulled directly from the CMS Cost Report:

- Average Length of Stay = Total Days / Total Discharges
- Occupancy Rate = Total Days / (Certified Bed Count * 365)
- Total Days
- Total Discharges
- Staffed Bed Count

Occupancy rate captures capacity utilization intensity and average length of stay reflects patient case complexity and operational efficiency. Total days captures overall utilization volume and demand burden while discharges and bed count provide scale context.

Structural Variable

The following structural variables were pulled from the CMS Provider of Services dataset:

- Rural vs. Urban designation
- Provider Category Subtype
- Provider Category Code
- State FIPS Code
- Program Participation Code

The above variables allowed the model to account for structural differences across hospitals such as geographic variation, rural vs. urban operating constraints, provider classification and regulatory status, and program participation that may influence payer mix and funding exposure. All of these variables were treated as categorical features and encoded using one-hot encoding within the preprocessing pipeline.

Preprocessing

All features were processed within a scikit-learn Pipeline:

- Median imputations for numeric variables
- Most frequent imputation for categorical variables
- Standard scaling for numeric features
- One-hot encoding for categorical features

We addressed class imbalance by using class_weight = "balanced" when applicable and by selecting evaluation metrics appropriate for rate event classification because financial distress is prevalent only 4-6% of the time.

Learning Methods Explored

We evaluate three fundamentally different supervised learning approaches:

Logistic Regression

Logistic regression acted as our baseline model as it estimates the probability of distress using a linear combination of features passed through a logistic link function. This model represents the probabilistic linear family of classifiers and we selected it for:

- Easily interpretable and understandable
- Provides calibrated probability outputs
- Performs well when relationships are approximately linear
- Allows us to assess whether complex models improve over a strong linear benchmark

Gradient Boosting Classifier

Gradient Boosting builds an additive ensemble of shallow decision trees, where each tree corrects the residual errors of the previous ensemble using gradient descent in function space. This model represents a tree-based method, which is fundamentally different from linear probabilistic models. We selected it for:

- Captures nonlinear relationships and feature interactions
- Handles heterogeneous feature types effectively after preprocessing
- It is robust to moderate multicollinearity
- Tree based methods often perform strongly in structured financial datasets

Support Vector Classifier

The Support Vector Classifier creates a maximum margin decision boundary by using a radial basis function kernel to project features into a higher dimensional space to capture nonlinear patterns. This model represents a non-probabilistic, margin based method which provides diversity in underlying learning mechanisms. We selected it fo:

- Uses margin based learning framework, distinct from probabilistic and tree based approaches
- Models complex nonlinear boundaries
- Offers different bias-variance tradeoff compared to boosting

Hyperparameter Tuning Strategy

For each model family, we did a systematic hyperparameter tuning using GridSearchCV with 5-fold cross validation on the training data (2017-2019). This allowed us to obtain stable performance estimates while still preventing enough data in the training split. Because this is a classification problem with class imbalance, the cross validation was stratified to maintain similar distress rates across folds.

Since financial distress is rare, we optimized models using Average Precisions (PR-AUC) rather than accuracy. PR-AUC focuses on performance for the distressed hospitals and reflects the goal of identifying at-risk facilities. We explored a reasonable range of hyperparameters for each model:

- Logistic Regression
    - Regularization strength (C) explored across multiple magnitudes to evaluate under and over regularization effects.
- Gradient Boosting Classifier - These parameters control model complexity, learning dynamics, and overfitting behavior.
    - Number estimator: 200-500
    - Learning rate: 0.01-0.1
    - Maximum tree depth: 2-3
    - Minimum samples per leaf: 1-30
- Support Vector Classifier - These control the tradeoff between margin width and misclassification, and the flexibility of the nonlinear decision boundary.
    - Regularization parameter ©
    - Kernel width: gamma

Hyperparameter combinations were selected based on mean cross validated PR-AUC. The best performing configuration within each model family was evaluated on the held out post pandemic test set (2022-2023) to assess forward generalization.

Summary of Methodological Approach

In summary, our supervised workflow was designed to prevent data leakage through careful feature exclusion and the use of a temporal train-test split, ensuring that the model does not rely on variables used to construct the distress label and that it is evaluated in a forward looking setting. We addressed class imbalance by selecting evaluation metrics appropriate for rare event prediction and by applying class weighting where applicable. To compare different modeling approaches, we evaluated three methods: Logistic Regression, Gradient Boosting, and Support Vector Classifier. For each model we used 5 fold cross validation to tune hyperparameters and select settings that balanced model flexibility with the ability to generalize new data. All preprocessing and modeling steps were implemented within reproducible pipelines. Together, this structured approach allows for fair model comparison and aligns that analysis with the real world objective of identifying hospitals at sustained financial risk.

Supervised Evaluation

Overall Results

Evaluation Metrics and Justification

Hospital financial distress is a rare event in our dataset with observations happening at 4-6% per year. Because of this, overall accuracy is not a useful metric. A model that predicts “not distressed” for every hospital would appear highly accurate but would fail entirely at identifying at-risk hospitals. Due to this we focused on:

- PR-AUC (Average Precision) was our primary metric. PR-AUC measures how well the model identifies distressed hospitals while balancing false positives. It is especially appropriate for rare-event classification problems.
- ROC-AUC was our secondary metric. ROC-AUC measures how well the model ranks distressed hospitals above non-distressed hospitals across all possible thresholds.

We used PR-AUC as the score metric during hyperparameter tuning and reported both PR-AUC and ROC-AUC for model comparison.

Cross Validated Comparison of Model Families

To compare models fairly, we report the mean performance across 5 fold cross validation, along with the standard deviation, for the best tuned model in each family. This prevents the results from being influenced by one specific train/test split. We also report performance on the test set (2022-2023) to assess performance on future data.

Model	CV ROC-AUC (std)	CV PR-AUC (std)	Test ROC-AUC	Test PR-AUC
Logistic (tuned)	0.014344	0.033861	0.737389	0.163762
Gradient Boosting (tuned)	0.007428	0.033486	0.886098	0.395271
SVC (tuned)	0.008859	0.040401	0.750646	0.175106

The above table shows that Gradient Boosting outperforms the other models specifically on PR-AUC. This suggests that nonlinear interactions and decision tree ensembles capture important structure in hospital financial data that linear and margin based models miss. Logistic regression provides a good baseline model and while SVC improves slightly it does not reach the same level of performance as Gradient Boosting. The small standard deviation across folds indicates that performance is stable and not very sensitive to particular splits of the training data.

In Depth Evaluation

We completed deeper analysis on Gradient Boosting as this achieved the strongest cross validation and test performance.

Feature Importance and Ablation Analysis

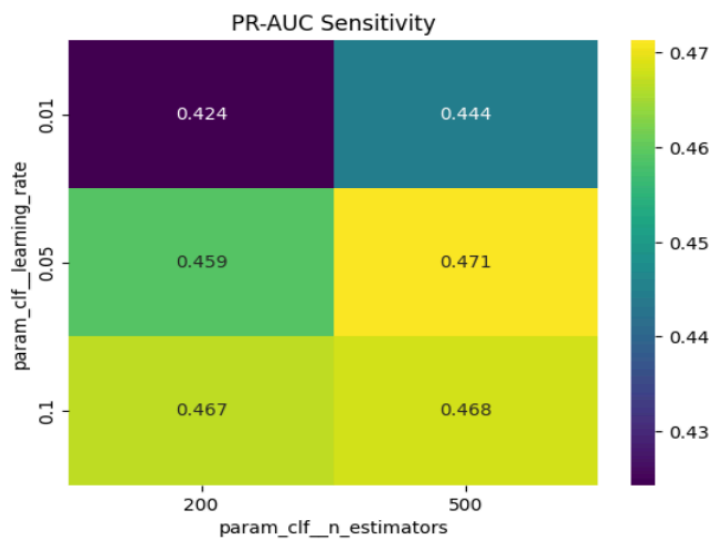
To understand which types of features contribute most to performance, we completed an ablation analysis. Rather than simply ranking features, we removed entire feature groups and measured how performance changed using 5 fold cross validation.

Model	CV ROC-AUC Mean	CV ROC-AUC STD	CV PR-AUC Mean	CV PR-AUC STD
GB All Features (tuned)	0.916881	0.007428	0.508627	0.033486
GB No Financial (tuned)	0.769923	0.010338	0.195570	0.030716
GB No Utilization (tuned)	0.910765	0.010005	0.478643	0.041206

By removing financial features, there was a large decline in PR-AUC mean from 0.5086 to 0.1956, indicating that financial ratios are the primary drivers of predictive performance. By removing utilization features, there was a small decline from 0.5086 to 0.4786, indicating that utilization metrics add good information to the model but are not the main source of predictive power. The ablation analysis confirms that combining financial, operational, and structural features produces better performance rather than relying on any single group alone.

Sensitivity Analysis

We examined how sensitive the tuned Gradient Boosting model was to key hyperparameters by reviewing cross validated PR-AUC across different combinations of learning rate and number of trees (Figure A). The heatmap shows how performance changes across these settings.



When the learning rate is set to 0.01, PR-AUC is noticeably lower and increasing it to 0.05 leads to a clear improvement in performance. However, moving from 0.05 to 0.10 does not change the results very much, as the scores remain close. Changing the number of trees from 200 to 500 has a smaller effect overall, with only modest differences in performance. The model's performance did not fluctuate dramatically across reasonable parameter choices. This indicates that the strong performance of the Gradient Boosting model is not dependent on a single narrow hyperparameter setting and is reasonably robust to tuning decisions. Overall the results indicate stable generalization rather than performance driven by overfitting to a specific configuration.

Identify Tradeoffs

Model performance also depends on the probability threshold used to classify hospitals as distressed. On the 2022-2023 test set we evaluated several thresholds:

Threshold	Precision	Recall	F1 Score	Proportion Flagged
0.05	0.270787	0.684015	0.387981	0.140931
0.10	0.349057	0.550186	0.427128	0.087939
0.20	0.444238	0.444238	0.444238	0.055792
0.50	0.565421	0.224907	0.321809	0.022192

Lower probability thresholds increase recall, meaning fewer distressed hospitals are missed but also result in more hospitals being flagged overall increasing false positives. In contrast, higher thresholds improve precision but miss a larger share of truly distressed hospitals. In our results, a threshold around 0.20 offers a more balanced outcome, with similar precision and recall and a proportion of flagged hospitals that is closer to the actual distress rate in the data. This reflects a practical decision point, increasing sensitivity helps catch more at risk hospitals, but it also requires more resources to review additional cases.

Failure Analysis

We analyzed prediction failure using the tuned Gradient Boosting model at the default 0.50 threshold. The confusion matrix on the test set shows True Negatives (9,012), False Positives (93), False Negatives (417), and True Positives (121). While the model performs very well on non-distressed hospitals, recall for distressed hospitals is lower at the 0.50 threshold.

Failure Categories

We grouped misclassified records into three main categories:

1. Missingness-heavy cases: hospitals with missing or zero utilization values that distort derived ratios.
2. Outlier financial ratios: hospitals with extreme ratio values outside the typical range observed during training.
3. Borderline probability cases: hospitals with predicted probabilities near the decision threshold.

Specific Failure Examples

- Provider CCN 230779 (2023): This hospital had zero inpatient days and zero discharges which lead to unstable occupancy and length of stay measures. Missing or non typical utilization data likely contributed to misclassification.
- Provider CCN 361308 (2022): This hospital had negative leverage, which is unusual and outside the normal training distribution, these outlier values can distort model predictions.
- Provider CCN 431302 (2023): This hospital had mixed signals, relatively high average length of stay but positive profitability. These conflicting indicators likely placed the hospital near the decision boundary.

Future improvements could focus on strengthening how the model handles irregular data patterns. For example, adding explicit missingness indicators may help the model better account for incomplete or zero utilization records. Adjusting how extreme values are handled could also improve the model. For example, limiting the impact of unusually high or low ratio values may prevent a small number of outliers from disproportionately influencing predictions. In addition choosing a probability threshold based on the intended use of the model, rather than defaulting to 0.50, could lead to more practical results. If the goal is to identify as many at-risk hospitals as possible, a lower threshold may be appropriate, while a higher threshold may be preferred if reducing false positives is more important.

Part B: Unsupervised Learning

Methods Description

Unsupervised Learning Workflow

Our goal in the unsupervised portion of the project was to identify natural groupings of hospitals based on financial and operational similarity, and then use those groupings to better understand patterns of hospital vulnerability, including whether certain clusters contained higher observed distress rates. To keep the clustering focused on more typical operating conditions, we limited the unsupervised dataset to pre-pandemic years (2017-2019) and then

aggregated to one row per hospital by taking hospital level averages across those years. This reduces noise from year to year fluctuations and avoids the major structural break introduced during COVID-era operations and funding. We implemented clustering in two iterations, Version A and Version B. Version A established a baseline using a smaller set of core ratios. Version B improved feature coverage and preprocessing after we observed that Version A clustering was being heavily influenced by extreme financial values.

Feature Representation and Preprocessing

Version A: Baseline Feature Set

We clustered hospitals using six standardized numeric features, operating margin, net margin, liquidity, leverage, average length of stay, and occupancy rate. The preprocessing steps were replacing infinite values with missing values, dropping hospitals missing any required features, and using Standard Scaler to put features on a comparable scale. We also ran PCA on the standardized feature mix to understand how much variance could be summarized in fewer dimensions and examine which features dominated the first components. PCA was not used as the clustering method but was used to diagnose feature dominance and support interpretation.

Version B: Expanded and Improved Feature Set

Based on Version A results, high silhouette values but clusters driven by extreme values, we expanded and refined the feature representation to better capture underlying hospital characteristics and reduce outlier influence. We expanded the feature set to include bead count, payor mix, care ratio, bed ratio, provider category, and urban vs rural indicator. The preprocessing steps were also more robust and included replacing infinite values with missing values, drop incomplete rows, clip numeric features at the 1st and 99th percentile to reduce extreme outliers, one-hot encode categorical features using dummy variables, and apply RobustScaler to numeric features as it is less sensitive to outliers. As in Version A, we used PCA for visualization and interpretability checks, not as the clustering algorithm itself.

Unsupervised Methods Explored

To meet the requirement for method diversity, we used two unsupervised clustering approaches with different underlying mechanisms. K-Means clustering assigns hospitals to clusters by minimizing within cluster distance to a cluster centroid. We used K-Means because it is easy to interpret as clusters have clear centers, it's scalable to thousands of hospitals, and a strong baseline for structured numeric data after scaling/encoding. Gaussian Mixture Model (GMM) assumes the data is generated by a mixture of Gaussian distributions and assigns hospitals to components based on probability. We used GMM because it allows for soft clustering membership as a hospital can be assigned with uncertainty, can capture elliptical cluster shapes that K-Means may not represent well, and provides model selection criteria that help choose the number of clusters. This gives us a fundamentally different clustering approach than centroid based K-Means.

Hyperparameter Exploration and Selection

K-Means: choosing the number of clusters (k)

For both Version A and Version B, we tested k=2 through 9 and selected the value of k that maximized the silhouette score. Silhouette score was chosen because if it reflects both it shows how close points are to their assigned cluster (cohesion) and how separated clusters are from each other (separation).

GMM: choosing the number of components (k)

For GMM, we tested 2 through 9 components and selected the best model using Bayesian Information Criterion (BIC) with AIC reported as a secondary reference. BIC was chosen because it balances model fit and complexity and helps prevent selecting an overly complex mixture model.

Explanation of Two Versions

Version A provided a clean baseline, but the combination of very high silhouette scores and PCA loadings suggested the clustering was being dominated by a few extreme financial values rather than broader hospital characteristics. Version B addressed this by expanding the feature set, applying outlier clipping and using more robust scaling, producing clusters that were more suitable for exploratory interpretation.

Unsupervised Evaluation

Overall Results Reports

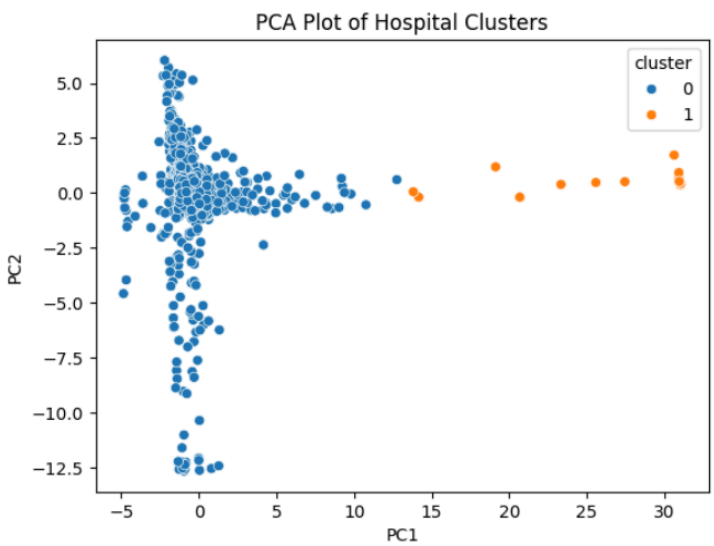
We used internal clustering metrics that align with each model's structure to evaluate and compare the two unsupervised methods.

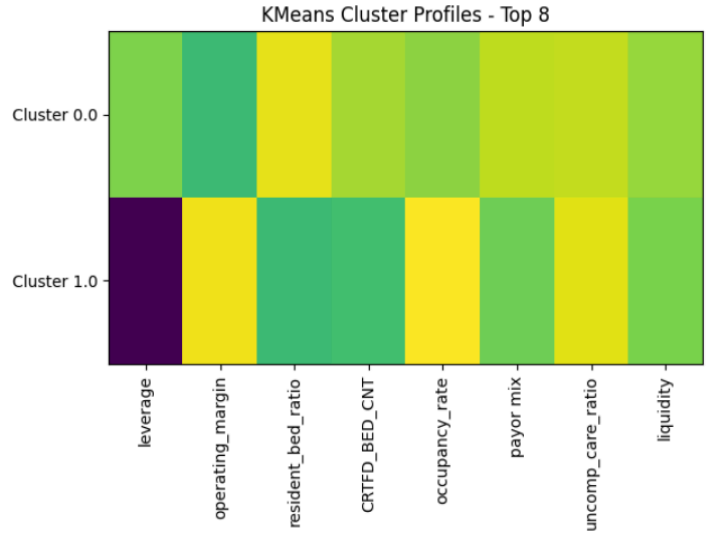
K-Means

For K-Means, we chose the number of clusters based on the Silhouette score, which reflects how closely hospitals resemble others in the same cluster compared to hospitals in different clusters. The best results occurred at k=2, with a Silhouetter score of 0.83433, suggesting strong separation between the two groups.

PCA Plot of Hospital Clusters (k=2): The PCA projection shows two separated groups along the first principal component. One cluster contains a small number of hospitals with extreme PC1 values, suggesting that this grouping is influenced by strong financial differences.

K-Means Cluster Profiles - Top 8 Features: The heatmap displays the top features that differentiate the two clusters. The differences can be seen the most in leverage, occupancy rate, resident bed ratio, uncompensated care ratio, and liquidity. This confirms that the clusters are driven by meaningful financial and operational characteristics rather than random variation.



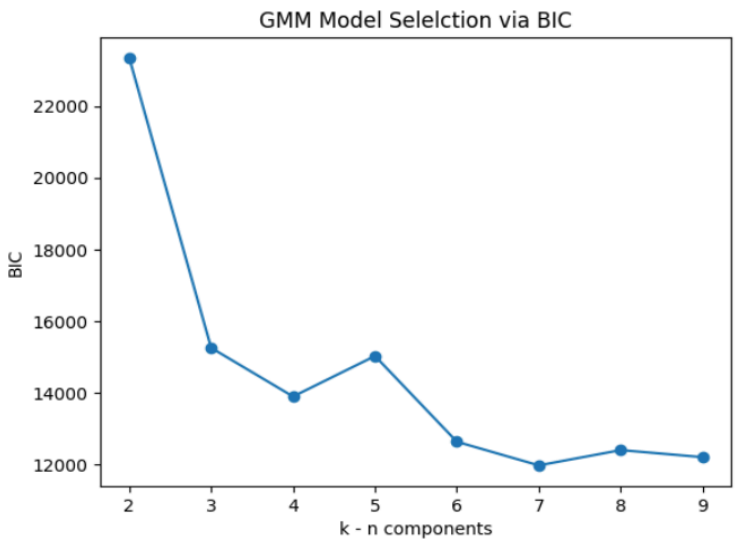


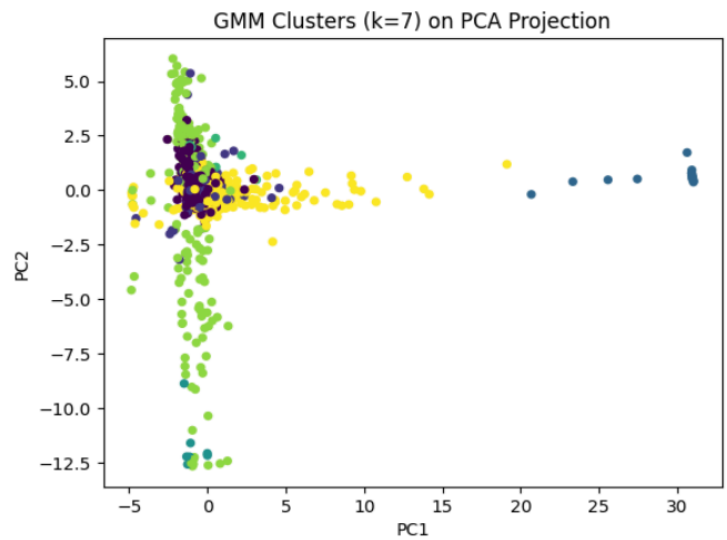
GMM

For GMM, we selected the number of components using the Bayesian Information Criterion (BIC), which balances model fit and complexity by penalizing models with more parameters. Lower BIC values indicate a better tradeoff between goodness of fit and overfitting. The lowest BIC occurred at k=7 using full covariance matrices (BIC = 11975.874, AIC = 10071.1005), indicating that seven mixture components provided the best fit under this framework.

GMM Model Selection via BIC: The BIC curve declines sharply from k=2 to k=3 and continues decreasing until reaching its minimum at k=7. After that point, improvements flatten. This pattern supports selecting seven components as the optimal balance between fit and complexity.

GMM Clusters (k=7) on PCA Projection: The PCA visualization of GMM assignments shows a more granular structure than K-Means. Instead of forming two groups, GMM identifies several overlapping subgroups, reflecting the probabilistic nature of the model.

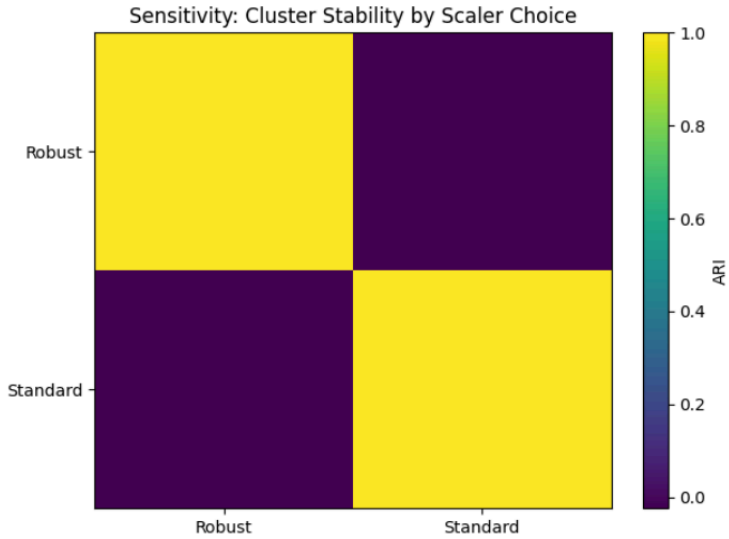




Sensitivity Analysis

In order to test how stable our clustering results were, we changed one model choice while keeping everything else the same. We replaced the RobustScaler with a StandardScaler and reran K-Means with k=2 and compared the two sets of cluster assignments using the Adjusted Rand Index (ARI). ARI measures how similar two clustering results are. A score of 1 means the groupings are identical, and a score near 0 means they are essentially unrelated. The ARI between the two scaling approaches was -0.024 indicating almost no agreement between the cluster assignments. In other words, simply changing how the data were scaled led to very different clusters.

The heatmap below shows this clearly. The diagonal value (1.0) represents perfect agreement when a method is compared to itself. The off-diagonal value near zero shows that switching the scaler changes which hospitals end up grouped together. This result tells us that the clusters are sensitive to preprocessing choices. Because unsupervised learning does not rely on labeled outcomes, small modeling decisions can significantly change the results. For that reason, we treat these clusters as exploratory patterns rather than fixed hospital categories.



Failure Analysis

We completed a structured failure analysis of our clustering results using the diagnostic outputs generated in our notebook. Based on our code generated categorization, most hospitals were labeled as stable, 939 records, meaning both K-Means and GMM produced consistent and confident assignments. However, a meaningful minority showed instability; 53 hospitals were labeled as method uncertain, 53 as outlier, and 9 as ambiguous boundary cases. This indicates that roughly 10% of hospitals do not fit cleanly into a single cluster structure.

The first failure pattern involves ambiguous boundary points. For example, Provider CCN 430077 was assigned to cluster 0 by K-Means and cluster 6 by GMM. Although its silhouette score was moderate, 0.625, it sat relatively far from the centroid, distance = 7.10, suggesting it lies near the edge of cluster space. These hospitals tend to share characteristics of multiple groups, and because clustering requires discrete assignments, they are forced into one cluster even if they naturally overlap categories. This reflects the limitation of hard clustering methods when boundaries are not sharply defined. The second failure pattern reflects method uncertainty, where the probabilistic GMM reveals low assignment confidence even when K-Means makes a clear decision. For instance, Provider CCN 220031 had a GMM maximum probability of only 0.60, indicating that its financial profile overlaps multiple mixture components. This kind of instability occurs when clusters are not well separated in feature space. While K-Means assumes spherical clusters and assigns membership necessarily, GMM exposes the underlying uncertainty. These cases highlight that cluster structure in hospital finance data is not always cleanly split. The third category includes outlier-like hospitals, such as Provider CCN 130006, which had a very low silhouette score, 0.019, and a large distance to centroid, 13.39. These hospitals have financial or operational profiles that are different substantially from typical members of their assigned cluster. Even after clipping extreme values and using RobustScaler in Version B, some hospitals remain structurally unusual. Clustering algorithms will still force these records into the nearest group, even if they do not meaningfully resemble it.

This failure analysis shows that our clustering model is better suited for exploring patterns then making firm classifications. Most hospitals group clearly, but a small number don't fit neatly due to overlap, uncertainty, or unusual profiles. In the future, we could flag extreme cases separately and add more time-based or structural data to make the clusters clearer and easier to interpret.

Discussion

Supervised Learning

One of the main takeaways from the supervised analysis is that nonlinear models effectively outperform linear models in predicting hospital financial distress. While Logistic Regression provided a solid and interpretable baseline, Gradient Boosting captured more complex interactions between financial ratios and operational variables, leading to higher PR-AUC. This reinforced the impression of testing multiple model families rather than relying on a single approach. We were surprised by how strongly financial ratios dominated performance in the ablation analysis. Removing financial features caused a dramatic decline in PR-AUC, while removing utilization features had a smaller effect. Although we expected financial metrics to matter, the magnitude of the difference confirmed that distress risk is primarily driven by financial structure rather than utilization patterns alone.

Data leakage was a challenge that we encountered. Initially performance appeared unusually high, which led us to re-examine the feature set. We realized that operating margin and liquidity, which were used to determine the distress label, were still included in the model. Removing those variables reduced performance but resulted in a much more realistic and defensible model. This step improved the integrity of the analysis and highlighted the importance of carefully reviewing feature construction in predictive modeling. Another challenge was handling class imbalance because distress is rare, standard accuracy metrics were misleading. Switching to PR-AUC and exploring probability thresholds allowed us to better understand practical tradeoffs between recall and precision.

With more time and resources, we would extend the supervised solution by incorporating additional years of data, adding time-series features such as financial trend changes rather than static averages, and exploring cost sensitive learning approaches that directly reflect policy or intervention priorities. We would also test external validation using a completely separate dataset if available.

Unsupervised Learning

We were surprised by just how unstable the cluster assignments could be. For example, when we tested the sensitivity of our best model by simply swapping our RobustScaler for a StandardScaler, the Adjusted Rand Index plummeted to negative 0.024. This negative score indicates total disagreement between the two methods, showing us that these clusters should be treated purely as exploratory tools rather than definitive risk categories. We were also surprised during our first iteration, Version A, when our principal component analysis and high silhouette scores revealed that the groupings were being completely dominated by extreme financial outliers rather than broad operational patterns.

Handling those extreme outliers was our biggest challenge. Because a few hospitals had massive financial anomalies, our initial KMeans model forced the data into skewed groups. This is what led to creating Version B of our model. In this improved version, we clipped all numeric features at the 1st and 99th percentiles and switched to a RobustScaler, which is less sensitive to extreme values. Another challenge we encountered was dealing with ambiguous cluster boundaries. Our failure analysis identified 53 hospitals that exhibited low confidence method uncertainty and another 53 that presented outlier like profiles. By categorizing these specific failures to acknowledge the limitations of our model, we concluded that while most hospitals form stable clusters, a small subset will always remain ambiguous.

If we had more time and resources, we would expand unsupervised learning by incorporating additional years of data and adding time series features. Right now, our unsupervised model uses static pre pandemic averages from 2017 to 2019. It would help to track longitudinal changes to see if a hospital shifts from a healthy cluster into a vulnerable cluster right before it experiences severe distress. Bringing in a completely separate dataset if available could help resolve the ambiguity for those uncertain hospitals.

Ethical Considerations

When we built our supervised learning model to predict financial distress, one major ethical issue we thought about was algorithmic bias. Since our model uses features like the urban versus rural indicator and program participation codes, it might be more accurate for certain types of hospitals than others. If policymakers actually used our model to decide who gets relief funding, a higher error rate for rural or safety net hospitals could mean vulnerable communities get unfairly denied the money they need to stay open. Another huge ethical issue is the cost of false negatives. If our model misses a truly distressed hospital, that community might lose its healthcare access without any warning. To address these issues, we would need to constantly audit the model to make sure its error rates are fair across all demographic and geographic groups. We would also need to work directly with health administrators to pick a probability threshold that leans toward catching at-risk facilities, and make sure human experts always review the model results before making any real funding decisions.

For our unsupervised clustering phase, the biggest ethical concern is the risk of accidentally stigmatizing hospitals. If a hospital gets grouped into a cluster that is seen as failing or high risk, creditors or state governments might panic and pull funding preemptively. By labeling a hospital as high risk, our model could accidentally scare off lenders and cause the very financial collapse we were trying to prevent. On top of that, our own sensitivity analysis proved that clustering is highly unstable. Just changing our scaling method from RobustScaler to StandardScaler completely scrambled the clusters, and our failure analysis found 53 facilities that the model was highly uncertain about. It would be incredibly unethical to attach real world consequences to such unstable groupings. To address this, we would have to be extremely clear with stakeholders about the model limitations. We would need to present the

clusters strictly as an exploratory tool to understand broad trends in the data, and warn against using them to label or penalize individual hospitals.

Statement of Work

Team members shared the responsibility for project design, interpretation, and written report.

- McKenzie Boutsikakis led data acquisition and preprocessing.
- Kostas Boutsikakis led supervised modeling and evaluation.
- Kirsten Emerson led unsupervised analysis and visualization.

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